

Unit V
Congruences and classical theorems

Course Objective – Cob5: To conceptualize congruence, its applications and standard theorems.

Course Outcome – CO5: To solve simple problems using classical theorems of number theory.

Bloom's Taxonomy Level: K1, K2, K3, K4

Part A

S.No.	Question	BTL*
1.	Prove that $a \equiv b(mod\ m)$ if and only if $a \equiv qm + b$	K2
2.	If $a \equiv b(mod\ m)$ and $c \equiv d(mod\ m)$ then show that I. $a \pm c \equiv b \pm d(mod\ m)$ II. $ac \equiv bd(mod\ m)$ III. $a^n \equiv b^n(mod\ m)$	K2
3.	If $a \equiv b(mod\ m)$ and c be any integer then show that i. $a \pm c \equiv b \pm c(mod\ m)$ ii. $ac \equiv bc(mod\ m)$	K2
4.	Solve $x \equiv 23^2(mod\ 5)$, if $3 \equiv 23(mod\ 5)$.	K2
5.	Using modular inverse, solve linear congruence $4x \equiv 7(mod\ 5)$	
6.	If it is 11.30am now, what time will it be in 1769 hours?	K2
7.	If today is Wednesday, what will it be in 129 days?	K2
8.	Find the last two digit in the decimal value of 1776^{1976}	
9.	State Chinese remainder theorem.	K1
10.	State Wilson's theorem with application	K1
11.	State Fermat's Little theorem	K2
12.	State Euler's theorem	K1

Part B

S.No.	Question	BTL *
1.	(i) Prove that congruence relation is equivalence relation. (ii) Prove that $a \equiv b(mod\ m)$ with $a > m$ and $b > m$ if and only if a and b leaves same remainder when they divided by m .	K4
2.	Using congruence relation, find the remainder when (a) 16^{53} is divided by 7. (b) 3^{247} is divided by 17.	K4
3.	Using congruence relation, (a) find the remainder when $1! + 2! + \dots + 100!$ is divisible by 15.	K4

	(b) find the remainder when $(n^2 + n + 41)^2$ is divisible by 12. (c) Find the positive integers 'n' for which $\sum_{k=1}^n k!$ is a square.	
4.	Find the last digit in the decimal value of (i) $1997^{1998^{1999}}$ (ii) $1943^{1642^{1053}}$	
5.	Using modular inverse, solve linear congruence (a) $3x \equiv 7 \pmod{31}$ (b) $5x \equiv 8 \pmod{37}$	K4
6.	Solve the following system of linear congruence (i). using Cramer's rule $3x + 13y = 8 \pmod{55}$, $5x + 21y = 34 \pmod{55}$. (ii). using Cramer's rule $2x + 3y = 4 \pmod{13}$, $3x + 4y = 5 \pmod{13}$.	K4
7.	Using chinese remainder theorem, solve the following Sun-Tzu puzzles a) $x \equiv 1 \pmod{3}$, $x \equiv 2 \pmod{5}$, $x \equiv 3 \pmod{7}$ b) $x \equiv 1 \pmod{3}$, $x \equiv 2 \pmod{4}$, $x \equiv 3 \pmod{5}$ c) $x \equiv 2 \pmod{3}$, $x \equiv 4 \pmod{5}$, $x \equiv 5 \pmod{7}$	K4
8.	Using Fermat's Little theorem, Find the remainder when a) 24^{1947} is divided by 17. b) 43^{5555} is divided by 31. c) 7^{1001} is divided by 17.	K4
9.	Using Fermat's theorem, solve linear congruence (a) $12x \equiv 6 \pmod{7}$ (b) $4x \equiv 2 \pmod{5}$	K4
10.	Using Euler's theorem, find the remainder when (a) 245^{1040} is divisible by 18. (b) 3^{247} is divided by 25	K4
11.	Using Euler's theorem, solve linear congruence (a) $35x \equiv 47 \pmod{24}$ (b) $25x \equiv 13 \pmod{18}$	K4